

Research Methodology

Protocol Registration

Guide to Systematic Literature Review

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What is Protocol Registration?

- A **protocol** is a pre-registered plan that outlines how you will conduct your SLR (objectives, search strategy, inclusion/exclusion criteria, data extraction, analysis plan, etc.).
- The goal is to make the process **transparent**, **reproducible**, and **bias-free**.

Do AI/ML SLRs *need* protocol registration?

- Unlike **medical or health sciences**, there is **no mandatory registry** (like PROSPERO) for computer science, AI, or data science SLRs.
- However, registering or publishing your protocol is **highly recommended** for transparency and credibility.



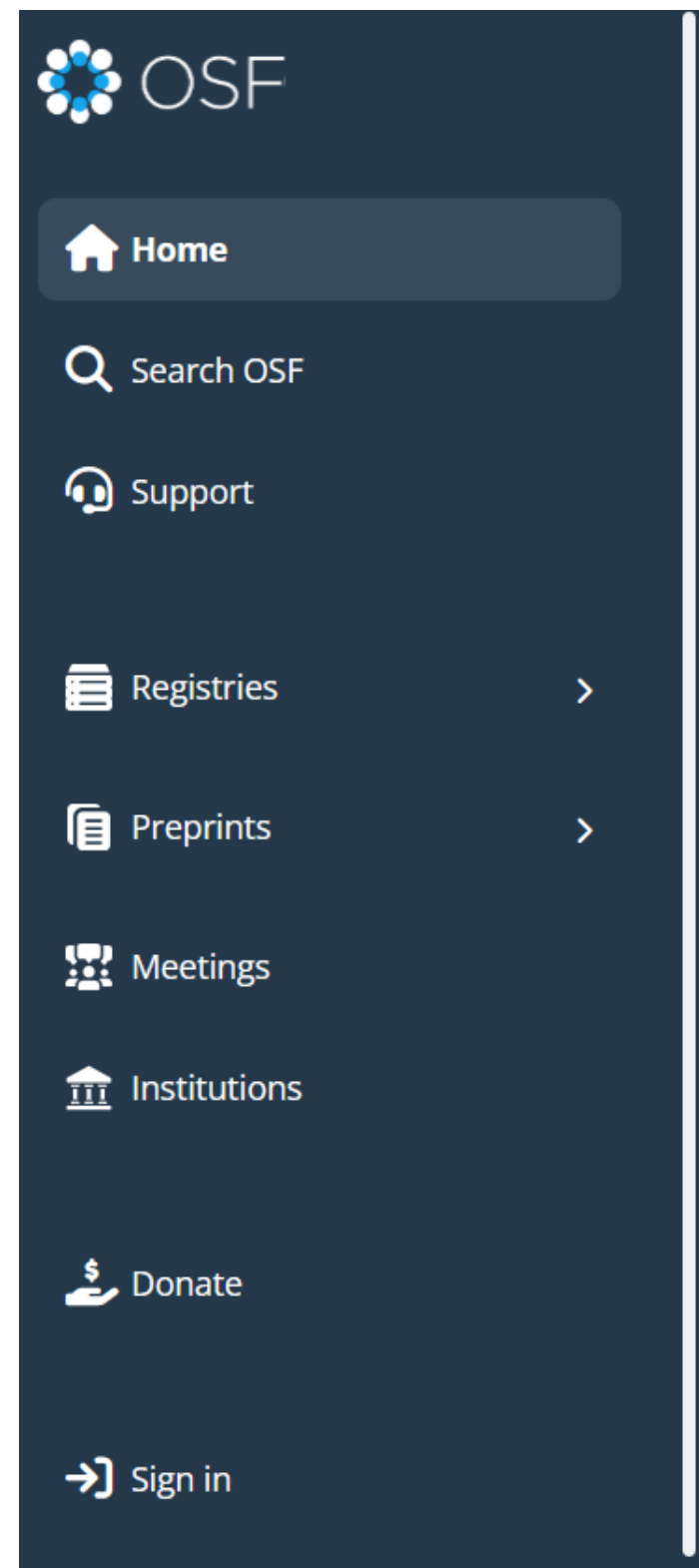
PROSPERO, OSF, ASPREDICTED



Option 1: Open Science Framework (OSF)

- **Most recommended for AI/Data Science SLRs**
 - Free and recognized for all research disciplines (not domain-specific).
 - You can upload your SLR protocol before starting data extraction.
 - Provides a **permanent DOI** and time-stamped record.
 - Easy to cite later in your published paper.
 - **Example structure for OSF protocol:**
- Title
 - Research questions (RQs)
 - Search databases (e.g., Scopus, IEEE Xplore, ACM Digital Library)
 - Inclusion/exclusion criteria
 - Quality assessment method
 - Data extraction process
 - Planned synthesis approach (e.g., thematic analysis, bibliometric mapping)
 - Register here: <https://osf.io/registries>





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Option 2: arXiv Preprint

- If you plan to submit your SLR to a top AI/ML journal (e.g., *ACM Computing Surveys*, *IEEE Access*, *Information Fusion*):
- You can **publish your protocol as a short preprint** on arXiv.
- This shows your methodology was planned in advance.
- Later, you can cite your preprint in the main SLR paper.



Option 3: Journal Protocol Paper

- Some computer science journals accept *protocol papers* these are detailed descriptions of your planned SLR **before** results are obtained.
- write your SLR without publishing a separate protocol.

Examples:

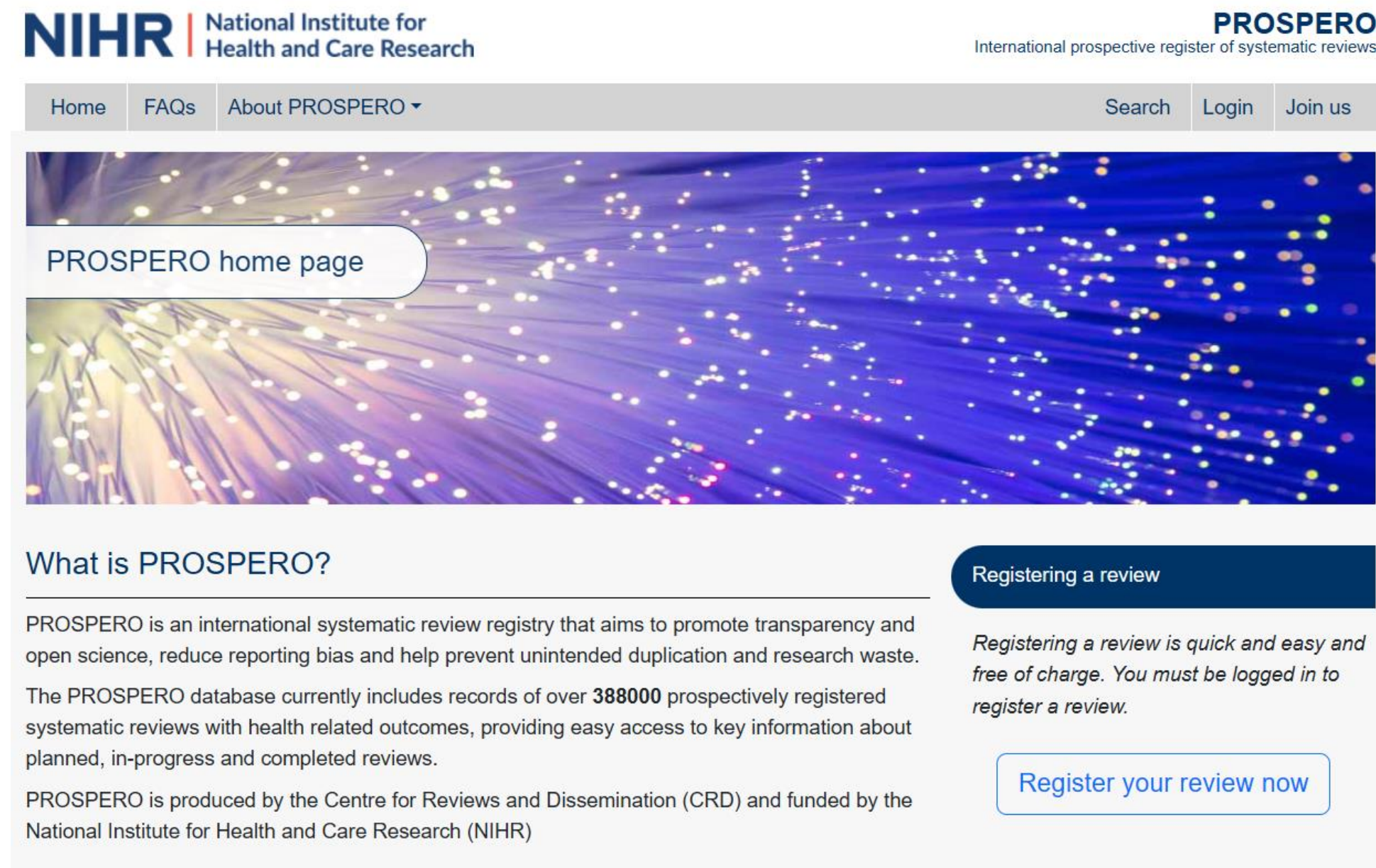
- *Environmental Modelling & Software*
- *Data & Knowledge Engineering*
- *MethodsX* (Elsevier, good for methodology protocols)
- However, this is optional you can directly



PROSPERO and AI/ML SLRs

- You **cannot** register AI/ML SLRs in PROSPERO.

That registry is **only for health-related systematic reviews** (e.g., interventions, clinical outcomes).



When to Register or Publish the Protocol

- You should register or upload the protocol **after defining your research questions and search strategy**, but **before** starting:
- Study selection
- Data extraction
- Analysis
- This ensures the SLR follows a **predefined, unbiased** plan.

Recommended Practice for AI/ML SLR Authors

- For credibility and easy citation:
- Prepare protocol document (in Word or Markdown).
- Register it on **OSF** with keywords and abstract.
- Add the registration DOI in your final SLR paper (e.g., “This review protocol is available at <https://osf.io/xxxxx>”).
- Optionally, upload a preprint on **arXiv** for visibility.

